

Engineering Document E16 — Infrastructure, Deployment & AI Gateway Architecture

Product Name: HQ

Engineering Package: Version 1.0

Purpose

This document defines the production infrastructure, deployment strategy, runtime environment, and AI Gateway architecture for HQ.

The infrastructure must support rapid MVP development while scaling to enterprise workloads without requiring major architectural changes.

Infrastructure should be invisible to users but reliable, secure, observable, and highly available.

Infrastructure Principles

The platform must be:

- Cloud Native
 - Containerized
 - Modular
 - Secure
 - Observable
 - Fault Tolerant
 - Horizontally Scalable
 - Cost Efficient
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Cloud Platform

Primary Cloud:

- Google Cloud Platform (GCP)

Core Services:

- Cloud Run (MVP)
- Cloud SQL (PostgreSQL)
- Cloud Storage
- Secret Manager
- Artifact Registry
- Cloud Logging
- Cloud Monitoring
- IAM

Future Scale:

- Google Kubernetes Engine (GKE)
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Runtime Architecture

Internet

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Cloud Load Balancer

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Next.js Web

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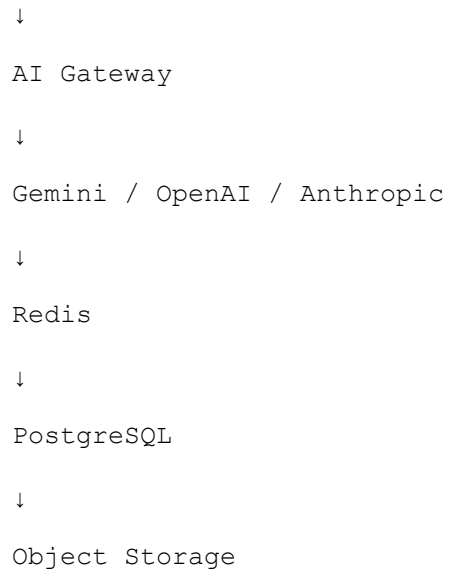
API Gateway

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NestJS Backend

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Mission Orchestrator



Each layer has a single responsibility.

Monorepo Deployment

Applications:

apps/
web/
mobile/
api/
workers/
packages/
ui/
sdk/
shared/
config/

Every application is independently deployable.

Containerization

Every service runs inside Docker.

Containers include:

- Web
- API
- Worker
- Scheduler
- AI Gateway

This guarantees consistent environments.

Environment Strategy

Supported environments:

Development

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Testing

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Staging

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Production

Each environment has isolated resources and secrets.

Environment Variables

Managed through Secret Manager.

Examples:

- Database URL
- Firebase Keys
- Gemini API Key
- OpenAI API Key
- Redis URL
- Storage Keys

Secrets are never committed to source control.

API Gateway

The API Gateway handles:

- Authentication
- Request validation
- Rate limiting
- Logging
- Routing
- Request IDs

All external traffic enters through the gateway.

AI Gateway

The AI Gateway abstracts all AI providers.

Mission

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AI Gateway

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Provider Router

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Gemini

OpenAI

Anthropic

Future Providers

Executives never communicate directly with providers.

AI Provider Selection

The gateway selects providers based on:

- Availability
- Cost
- Model capability
- Latency
- Organization preference
- Mission requirements

Routing rules are configurable.

Failover Strategy

If a provider becomes unavailable:

Gemini

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Retry

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OpenAI



Anthropic



Mission Retry



CEO Notification

Service continuity is prioritized.

Background Workers

Workers execute:

- Long-running missions
- Asset generation
- Email delivery
- Notifications
- Document processing
- Analytics aggregation

Workers operate independently of user requests.

Queue System

Redis queues manage:

- Mission execution
- Executive tasks
- Email jobs
- AI requests
- File processing

Queue priorities include:

Critical

High

Normal

Low

Scheduler

Scheduled tasks include:

- Subscription renewal
- Memory cleanup
- Analytics aggregation
- Health checks
- Daily reports
- Organization maintenance

All scheduled work is centralized.

Database Infrastructure

Primary Database:

PostgreSQL

ORM:

Prisma

Readiness:

- Automatic migrations
- Backups
- Recovery

- Monitoring
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Object Storage

Stores:

- Uploaded assets
- AI-generated images
- Documents
- Videos
- Reports
- Exports

Storage is independent of compute resources.

Caching

Redis caches:

- User sessions
- Organization settings
- Executive metadata
- Prompt modules
- Frequently accessed analytics

Caching reduces latency and database load.

Search Infrastructure

Future support:

- Full-text search
- Semantic search
- Knowledge indexing
- Mission search

- Asset search

Search integrates with the Memory Engine.

Logging

Centralized logging captures:

- API requests
- AI executions
- Executive actions
- Errors
- Queue events
- Authentication
- Billing events

Every request receives a unique Request ID.

Monitoring

Monitor:

- CPU
- Memory
- Response time
- Queue length
- AI latency
- Error rates
- Database performance
- Cache performance

Dashboards provide real-time visibility.

Health Checks

Every service exposes:

- Liveness endpoint
- Readiness endpoint
- Dependency status

The platform automatically detects unhealthy services.

Security

Infrastructure security includes:

- HTTPS everywhere
- Encryption at rest
- Encryption in transit
- Secret management
- IAM roles
- Firewall rules
- Network isolation

Security is enforced at every layer.

CI/CD Pipeline

Pipeline stages:

Code

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Lint

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Test

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Build

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Docker Image

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Deploy to Staging

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Integration Tests

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Deploy to Production

Deployment is automated and repeatable.

Backup & Recovery

Automated backups include:

- PostgreSQL
- Object Storage metadata
- Configuration
- Audit logs

Recovery procedures are tested regularly.

Scalability Strategy

Horizontal scaling applies to:

- API
- Workers

- AI Gateway
- Web Frontend

Stateless services enable rapid scaling.

Cost Optimization

Strategies include:

- Autoscaling
- Idle instance reduction
- Intelligent provider routing
- Cache optimization
- Storage lifecycle policies

The platform balances performance and operational cost.

Disaster Recovery

Recovery objectives:

- Automated restart
- Database restoration
- Queue recovery
- Object restoration
- Configuration recovery

Critical services are prioritized.

Future Evolution

The architecture supports:

- Multi-region deployment
- Edge computing

- Enterprise private deployments
- Hybrid cloud
- On-premise installations
- Additional AI providers

Growth should not require redesigning the infrastructure.

Success Criteria

The Infrastructure & AI Gateway Architecture succeeds when:

- Services deploy consistently across environments.
 - AI providers are abstracted behind a unified gateway.
 - Background processing scales independently.
 - Infrastructure is secure, observable, and resilient.
 - The platform can grow from hackathon MVP to enterprise deployment with minimal architectural change.
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Conclusion

The HQ Infrastructure, Deployment & AI Gateway Architecture establishes a cloud-native operational foundation for the platform.

By combining containerized services, Google Cloud infrastructure, an AI Gateway, background workers, Redis queues, PostgreSQL, automated deployment pipelines, and comprehensive monitoring, HQ gains a resilient and scalable runtime capable of supporting sophisticated multi-agent AI workflows today while remaining prepared for future enterprise expansion.